

**Faculty of Business Informatics
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Face Recognition using Machine Learning

Bachelor Thesis

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This is to certify that:

- (i) the thesis comprises only my original work towards the Bachelor degree,
- (ii) due acknowledgement has been made in the text to all other material used.

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May 17th, 2026

Abstract

This thesis focuses on improving face recognition performance under uncontrolled conditions, where factors such as pose variation, occlusion, and appearance changes can reduce recognition accuracy. To address these challenges, a face recognition system was developed using FaceNet for feature extraction and a Support Vector Machine for classification within a Design Science Research framework. The proposed system also incorporates data augmentation and a hybrid decision strategy to improve recognition reliability and distinguish between known and unknown users. Experimental results achieved high accuracy, precision, recall, and F1-score values, demonstrating the effectiveness of the proposed approach under the tested conditions.

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List of Acronyms and Initialisms

ANN	Artificial Neural Network
CNN	Convolutional Neural Network
DBN	Deep Belief Network
DL	Deep Learning
DSR	Design Science Research
DSRM	Design Science Research Methodology
FAR	False Acceptance Rate
FN	False Negative
FP	False Positive
FR	Face Recognition
FRAD	Face Recognition and Detection
FRR	False Rejection Rate
KNN	K-Nearest Neighbors
LBP	Local Binary Patterns
LFW	Labeled Faces in the Wild
LS-SVM	Least Squares Support Vector Machine
ML	Machine Learning
PCA	Principal Component Analysis
RBF	Radial Basis Function
SVM	Support Vector Machine
TN	True Negative
TP	True Positive

Chapter 1

Introduction

This chapter provides a brief introduction to the seminar, including the basis for it and the business challenge it aims to answer. This chapter also provides a breakdown of the contributions done in this seminar along with an explanation on how it's organized.

1.1 Motivation

Face Recognition (FR) is one of the most distinguished cognitive abilities that researchers have been putting great effort into replicating through a large number of complex algorithms and uniquely insightful methodologies. The core of FR systems relies on effectively recognizing multiple significant features such as nose structure, eye socket placement, jaw prominence, forehead shape, inter-eye distance, and cheekbones etc [1]. Face Recognition revolves around identifying “to whom a given face or image belongs.” The process typically involves four key stages: detection, feature extraction, tracking, and recognition [2]. Together, these stages enable an automated system to analyze, encode, and match facial informa-

tion for identification or verification.

One of the most commonly used approaches for FR is machine learning (ML), particularly deep learning (DL) methods, which can extract more complex facial characteristics. Deep learning has made significant progress in overcoming challenges that have long restricted the capabilities of the artificial intelligence community [3]. However, despite these advancements, several challenges still persist. Face detection of pictures is usually complicated because of various human facial features such as position, orientation, expression, and skin color [2]. Earlier FR algorithms were trained on images captured in controlled environments, where lighting, angles, and facial expressions were carefully managed. As a result, their accuracy drops significantly under real-world conditions such as low resolution, blur, pose variation, and occlusion. Most image-based data today come from low-resolution devices, which makes achieving high accuracy even more challenging [4].

Although extensive research has been conducted in this field, achieving truly robust and reliable FR systems that can perform effectively across diverse real-world conditions remains an unsolved challenge [5]. Therefore, this paper proposes an improved FR approach that can address these limitations and enhance performance in uncontrolled environments.

1.2 Contributions

This paper proposes an enhanced FR model designed to perform effectively under uncontrolled environmental conditions such as varying lighting, pose, occlusion, and low resolution. The system leverages multiple ML techniques to optimize

feature extraction and classification accuracy, enabling robust recognition in real-world, non-ideal scenarios. Additionally, the model explores the integration of preprocessing and adaptive learning techniques to minimize the negative effects of visual inconsistencies and improve overall stability.

1.3 Organization

This paper is organized as follows. Chapter 1 provides an introduction to the thesis, outlining the motivation, objectives, contributions, and overall structure of the paper. Chapter 2 presents the background and literature review, covering key concepts related to FR, its applications, and the factors affecting its performance. It further explores various ML techniques, commonly used algorithms, their strengths and limitations, as well as how these approaches are applied to FR. The chapter concludes with a discussion on datasets, evaluation criteria, current challenges, and future trends in the field.

Chapter 3 details the proposed research methodology, presenting the identified research gap, problem statement, and research questions and aims. It also explains the design science framework adopted in the thesis, including its types, and outlines the potential tools and software packages utilized. The chapter concludes with a summary of the research design and implementation approach.

Chapter 2

Background and Literature Review

In this chapter, a summary is presented of the issue that this research seeks to tackle, namely enhancing the effectiveness of FR systems in practical situations. Face Recognition has gained significant importance because of its common use in security measures, authentication processes, and biometric verification systems. In addition to this, the chapter explores how ML, especially DL, contributes to improving the effectiveness of FR [4].

This chapter also discusses elements that influence the precision and dependability of FR, including pose, lighting, obstruction, and image resolution. Additionally, it examines the datasets that are frequently utilized to train and assess FR models, along with the assessment standards used to determine their efficiency.

2.1 Face Recognition

Identifying individuals has become increasingly common through biometric identification in recent times. Instead of using passwords, PINs, smart cards, plastic cards, tokens, or keys to verify who someone is and give them access to real or digital spaces, these techniques focus on an individual's physical traits or behavior to establish their identity [6].

Among the various biometric technologies, FR is one of the most popular and extensively studied. This process involves automatically identifying or confirming a person's identity by analyzing their facial features. Face recognition is a highly accurate and non-invasive method that merges the advantages of physical methods with privacy [7].

Furthermore, FR is a crucial component in distinguishing between faces once they have undergone grayscale processing. Face Recognition involves identifying the face and checking it against certain images for verification or identification. The captured face image is used as input and is compared with images stored in a database to confirm its identity. If the features of the detected face match with those in the database, then the identification is successful [4]. The wide-ranging applications of FR have sparked considerable interest in the technology, leading to the emerging demand for development of robust algorithms capable of dealing with real-world facial images [5].

2.1.1 Applications of Face Recognition

As we are aware, continuous work is being done to create a FR system that is highly accurate, mainly due to its importance in various fields. Here, a few of its uses are highlighted, such as video surveillance, access control, security and pervasive computing.

Surveillance

Facial recognition technology places a strong emphasis on surveillance because video offers a wealth of data for identifying people [8]. In contrast to other biometrics like iris or finger recognition, FR is useful for identification at a distance without requiring the subject's active involvement or knowledge [9]. Face Recognition may be used with conventional cameras to effectively record faces in large public spaces, where it is frequently necessary to identify people at distances of 10 to 20 meters or more for security, defense, and commercial reasons [9].

Face Recognition can be incorporated into CCTV systems to identify persons on watch lists, locate missing people, notify law enforcement agencies of known criminals or drug offenders, and more. Using smart camera networks, Face Recognition at a Distance (FRAD) enables long-range identification, whitelist verification, and re-identification across vast regions [10, 9].

The term "surveillance" comes from the French words *sur* ("from above") and *veiller* ("to watch"), and it entails monitoring individuals, actions, or associated data for safety and security. Electronic devices like CCTV cameras or data interception can be used to accomplish this. Surveillance is useful for gathering intelligence, preventing crimes, and monitoring processes, individuals, or groups,

but it is also subject to criticism for possible privacy breaches, which has prompted many democracies to legislate restrictions on its use. Surveillance methods are generally similar across countries, and ongoing research is constantly enhancing FR algorithms to produce more trustworthy outcomes [11].

Despite the fact that FR-based surveillance systems have begun deployment, their current performance is still inadequate for use in all large-scale projects. Further advancements and investments, such as non-visible wavelength imaging, steerable zoom cameras, and sophisticated signal processing, are anticipated to enhance future capabilities [8].

Access Control

Face recognition is commonly used to control access to systems like computers, buildings, offices, ATMs, airports, and seaports by first confirming the identity of authorized individuals before granting access or use of services [12]. When there are few users and the stored face images are captured under regulated conditions, these systems achieve high accuracy, which lowers the need for user participation during authentication [12]. Modern access control systems have the capability of constantly monitoring who is using a device and implementing security measures when the user leaves, such as locking the terminal until the right individual is identified once more [12]. Face recognition can also manage access on ATMs or personal computers by making sure that only authorized users are allowed to reactivate the system after it has been left unattended, hence securing private accounts and sensitive data [11]. Facial recognition may be used in banking applications to replace or supplement ATM cards and passwords by taking a real-time image of the user and comparing it to the image that the bank has saved in its database to

grant access [12].

Security

In many settings, security is a primary priority, and FR enhances it by aiding in identity verification and investigative activities like licensed driver authentication, tracking missing people, immigration inspections, and general identity verification, including national IDs, passports, banking, and employee identification [11]. Face recognition systems can also be used in banks, airports, schools, offices, and other places where it's important to restrict unauthorized access [12]. Compared to many other biometric techniques, it may provide better security for computer systems by making sure that only authenticated users are allowed to log in or change data [12].

Pervasive Computing

The integration of microprocessors and sensors into everyday devices, allowing them to gather, analyze, and send data to enhance quality of life in sectors like healthcare, safety, logistics, and consumer technology, is known as pervasive computing [11]. It employs intelligent sensor networks, RFID tags, wearable devices, embedded systems, and wireless communication to build environments that are aware of their surroundings and react to user needs [11].

Facial recognition is regarded as a good biometric technique for pervasive computing since it is non-intrusive and enables identification without the need for physical contact or user interaction [8]. In pervasive settings, a primary objective is to enable devices to identify the people in their immediate vicinity, which

facilitates tailored responses and more intuitive human-computer interaction [8]. Integrating face identification enables devices to function with more awareness and automation, as many of them already have cameras and sensors but continue to rely heavily on manual input [12].

Facial recognition can be used by pervasive systems to identify users for personalization, regulate access to services, and ascertain who is present in an environment in order to facilitate context-aware capabilities such as message routing or system behavior modification [8]. Modern devices, such as smartwatches that handle alerts and communication, illustrate how sensor-based systems improve everyday interactions and reflect the overarching goals of pervasive computing [11]. It is anticipated that facial recognition will become more and more crucial in facilitating smart and adaptive device behavior as these technologies develop [8].

2.1.2 Factors Affecting Performance

A covariate is a factor that causes variation within classes to increase, reduces variation between classes, or impacts both aspects. Covariates that can be regulated during the process of image collection signify the limitations of facial recognition technology. For example pose, illumination, expression, and image resolution. The challenges posed by these covariates have been thoroughly examined, leading to the creation of many algorithms designed to tackle them [13].

In contrast, uncontrollable covariates such as race, gender, age groups, and aging also affect the performance of FR algorithms. Notably, aging has garnered significant focus in research, and specific databases have been established to assist scientists in addressing this issue [13].

Controlled Covariates

Controlled covariates are the attributes of an image that can be adjusted or influenced by the user in a designated environment or situation. For example, when a person approaches an automated passport control gate, a picture is taken under regulated conditions, usually when the individual faces the camera squarely and maintains a neutral expression [13].

Pose Variation

Pose variation is one of the major challenges in achieving a successful FR system. It occurs during image acquisition when a person is not directly facing or looking straight at the camera, resulting in different facial angles and positions in the captured image [13]. As a result, certain facial features may become invisible or distorted, which significantly reduces the accuracy of FR algorithms. In fact, the differences between images of the same person taken at varying poses can sometimes be greater than the differences between two different individuals, making recognition even more difficult [13].

Since there is no standardized or fixed way for people to pose, these variations create inconsistencies in facial features and degrade overall system performance. Many recognition systems also operate under inflexible imaging conditions, which further affects the quality of gallery images and limits their ability to handle pose changes effectively [11].

Researchers have proposed various approaches to deal with this issue. Broadly, the methods used to address pose variation can be divided into two main categories:

multi-view FR and FR across pose. In multi-view FR, the gallery contains multiple images of each person captured from different angles, allowing the system to reference these views during recognition. In contrast, across-pose recognition involves identifying a face presented in a pose that the system has never encountered before, making it a more complex task [11].

A good FR approach should be capable of recognizing faces from multiple viewpoints and exhibit strong tolerance to pose variation. However, despite extensive research, achieving complete accuracy in such systems remains an open challenge. Most algorithms still struggle when faces are captured from extreme or unfamiliar angles [11].

Illumination

The visible attributes and effects of light are known as illumination, and they have a big impact on how a face looks in a picture [11]. Variations in lighting, such as changes in intensity, direction, or the distribution of light sources, can significantly affect a face’s visual appearance, producing shadows and shading that may cause the same person to seem very different under various conditions [5, 13]. The accuracy of FR systems can be diminished by these variations, which can result in differences that are even greater than those between different people [13]. In uncontrolled environments, where elements like skin reflectance, camera sensor response, and internal camera controls have a greater impact on image quality, the problem of illumination is exacerbated [5].

Identifying individuals from video or still photographs under varying lighting conditions is challenging. When images are acquired in regulated environments with

consistent lighting and background, recognition is often more trustworthy [11]. Researchers have created a number of strategies to deal with variations in lighting, such as gradient extraction, gray-level transformation, and face reflection estimation [11]. Gray-level transformation performs detailed mapping utilizing linear or nonlinear functions to lessen the effects of illumination changes, while gradient extraction concentrates on identifying edges from gray-level images [11]. Many current approaches, however, depend on stringent assumptions, such as needing several photos of the same individual under different lighting circumstances, which are often unrealistic [5]. Although more sophisticated approaches like picture re-lighting and albedo estimation have been created to produce several frontal images under varying lighting conditions, these techniques continue to have issues with image alignment and sensitivity to facial expressions [5].

Illumination variation is still one of the main technical obstacles to creating FR systems that are both reliable and efficient. Ongoing research is still focused on finding better ways to deal with these variations in both regulated and uncontrolled environments [13].

Partial Occlusion

In face identification, partial occlusion happens when a face is partially obscured in an image by either natural or artificial objects. If fewer than 50% of the face is obscured, such occlusions are typically regarded as partial. Scarves, hats, hair, hands, sunglasses, and other things that hide the face can all be included in this category [11]. Occlusion poses a major problem since it hides essential facial features, making identification either harder or less precise. When occlusions are deliberate, such as when individuals use items or wear accessories to hide their

face, the issue is more pronounced. The recognition process might be made even more complex by occluding factors, such as severe illumination-induced shadows [11].

To combat partial occlusion, researchers have devised a number of methods. Local approaches concentrate on breaking the face down into distinct areas and analyzing the characteristics of each section, identifying the best potential features, and combining them for identification. In contrast, near holistic approaches get rid of the hidden features and make use of the remaining visible components of the face as useful data for identification [11]. Generally, partial occlusion recognition strategies fall into three categories: Fractal-Based Methods, Feature-Based Methods, and Part-Based Methods, each of which takes a somewhat different approach to the problem [11].

Due to its prevalence and impact on system performance, partial occlusion has attracted a lot of attention in recent years. One of the major problems in automated face identification systems is enhancing recognition in these situations, and several approaches are constantly being created to do so [13].

Uncontrolled Covariates

A person's face has features that cannot be controlled or altered during image capturing, and these are known as uncontrolled covariates. These include inherent features such as age group, race, gender, and the effects of aging [13]. Uncontrolled covariates, unlike controlled covariates, which may be standardized through the imaging procedure, create variability that has a major impact on the performance of FR systems. Because these variables are intrinsic to the individual, they pose

persistent difficulties for algorithm developers, necessitating specialized datasets and approaches to account for the changes they produce in facial expression [13].

Aging

Face identification systems are severely affected by aging, which is an unavoidable and uncontrollable process [11]. It manifests in three primary ways: it is slow and irreversible, it exhibits personalized patterns based on a person’s genes, health, diet, geography, and environmental variables, and the visible signs of aging are time-dependent, affecting older faces more significantly than younger ones [11]. As people get older, there are significant differences in facial appearance within a class, including wrinkles and structural changes in facial shape [12, 13]. In fact, these modifications can alter discriminatory facial features, making it extremely difficult to recognize people over significant age gaps [13]. However, it is challenging to gather data for training FR systems because aging is a gradual process [12]. As a result, FR systems must be able to account for the effects of aging. Despite the fact that the majority of studies agree that identification performance declines with increasing age differences, this decline is more noticeable over years than months or days [13]. Although current research focuses on age estimation, age simulation, and cross-age recognition, limited datasets continue to restrict progress [13].

Gender

Due to variations in local facial features and forms between men and women, gender has an impact on face identification. Generally speaking, women have smaller noses and smoother cheeks, while men have stronger jaw structures. Although it can be difficult for children to distinguish between genders due to similarity in

early skeletal structures, anthropological research confirms that male and female facial structures differ significantly [13].

Race

Race refers to biological and physical characteristics such as skin color, hair, eye, and bone structure. Race is especially important in FR since differences in facial features and skin tone among populations such as African, Caucasian, and East Asian groups can significantly affect algorithm accuracy [13].

2.2 Machine Learning

The field of computer science known as ML is dedicated to creating systems that can learn and improve on their own using data and experience. It tackles two interconnected issues: what basic computational, statistical, and information-theoretic concepts regulate learning in both people and machines, and how to create systems that learn without specific programming [14]. In the last 20 years, ML has gone from being a theoretical curiosity to a useful technology used in a variety of sectors and scientific study. It is now the method of choice for artificial intelligence applications, where training a system on examples of desired input-output behavior, has proven to be more effective than manually programming responses in areas like robot control, natural language processing, speech recognition, and computer vision [14].

The fundamental idea behind ML is that it allows computers to learn from data and thereby enhance their performance on certain tasks over time [15]. Although the distinction between ML and conventional statistical methodologies might be

hazy, ML generally places a high premium on prediction accuracy and frequently deals with massive, high-dimensional datasets that traditional methods may have difficulty analyzing [15]. The field has had a significant influence on a variety of data-intensive fields, including biology, cosmology, social sciences, fault diagnostics, logistics, consumer services, computer science, and technology [14]. Its increasing significance is especially apparent in the age of Big Data, where ML techniques shine at identifying useful patterns from massive, complicated datasets [15]. The study of ML provides researchers with the skills necessary to evaluate, implement, and innovate in a variety of scientific and practical domains [14, 15].

2.2.1 Types of Machine Learning Algorithms

Machine learning methods are broadly classified into four learning methods: Supervised learning, Unsupervised learning, Semi-supervised learning, and Reinforcement learning. The paper discusses these approaches one by one in detail below.

Supervised Learning

For each observation, the outcome or dependent variable, sometimes known as the label, is already known in supervised learning, a ML technique that uses labeled data [15]. This method of learning involves splitting the dataset into training and testing sets, which enables the algorithm to identify patterns in the training data and then make predictions or decisions on novel, unobserved data [3]. The main objective is to use sample input-output pairs to train a function that maps inputs to outputs, allowing the model to estimate the link between features and results [16].

Since it operates with particular goals that are defined by given inputs, it is said to be task-oriented [16]. Regression, which deals with continuous-value challenges, and classification, which solves discrete-value problems, are the main uses of supervised learning [3]. Linear regression, logistic regression, decision trees, and support vector machines(SVMs) are some of the most popular supervised learning techniques [15]. An example of this method is text classification, where the model predicts the class label or sentiment of a text such as a tweet or product review [16].

Unsupervised Learning

The data-driven process of unsupervised learning, a kind of ML, analyzes unlabeled datasets without the need for specific output labels or supervision [16]. Without considering a "right answer" or target outcome, this method allows the algorithm to find natural patterns, relationships, or groupings within the data [15]. It teaches the model to identify underlying patterns or configurations using the characteristics of unlabeled data [3]. Clustering, dimensionality reduction, and anomaly detection are the most typical unsupervised learning activities [16]. Observations with comparable features are often grouped using clustering methods such as k-means and expectation-maximization employing Gaussian mixture models [15].

Semi-Supervised Learning

In semi-supervised learning, a hybrid methodology is used to create models or classifiers by combining both labeled and unlabeled data [17, 16]. It sits between supervised and unsupervised learning, employing a small amount of labeled data

and a large amount of unlabeled data to increase prediction accuracy [16]. In actual settings where unlabeled data are plentiful but labeling data is expensive and labor-intensive, this technique is especially helpful [15]. Including unlabeled data can improve model performance, according to research. For example, a naive Bayes classifier that was trained on both labeled and unlabeled data performed 30% better in accuracy in a web page classification task than one that was just trained on labeled data [15]. Applications of semi-supervised learning include fraud detection, text classification, and machine translation [16].

Reinforcement Learning

Reinforcement learning is a kind of ML in which systems learn to execute tasks through trial and error, gradually improving with time based on feedback from the environment [15]. Because the model performs actions within an environment and receives either rewards or punishments depending on the results of those actions, this approach operates dynamically [16]. Through this iterative process, the algorithm self-adapts, refining its behavior to maximize rewards or minimize risks [15]. Depending on how it is implemented, reinforcement learning can be similar to supervised or semi-supervised learning, such as in generative adversarial networks [15]. It is extensively employed in gaming, robotics, automation, and navigation systems, where agents are constantly learning the best course of action in complicated and ever-changing situations [3, 16].

2.2.2 Algorithms Commonly Used

In this section, we discuss various ML algorithms such as SVMs, Naïve Bayes, K Nearest Neighbors (KNN), Neural networks and K-means clustering.

Support Vector Machines

Support Vector Machines are a type of supervised learning approach that may be used for regression and classification jobs [16]. Vladimir Vapnik first introduced them in 1965 [11]. Support Vector Machines aim to determine the optimum hyperplane that separates classes in linearly separable data using the decision function $f(x) = w^*.x + b^*$, where w is a weight vector, b is the bias, and x is the problem variable. We only need the sign of the decision function to ascertain the class to which a data item x_i belongs [11]. The hyperplane chosen by SVM is the one that maximizes the margin, which is the space between the hyperplane and the closest points to it, also known as support vectors [11]. A greater margin typically translates into better generalization [16]. Generally, SVMs have a low rate of misclassification and do well with high-dimensional data. SVMs possess a reasonable level of interpretability, particularly when a kernel function is not used [15].

Kernel functions are used by SVMs for nonlinear problems [11]. Typical kernels include linear, polynomial, radial basis function (RBF), and sigmoid [16]. The employment of a sigmoid kernel in an SVM model is akin to a two-layer perceptron, which has strong ties to neural networks [17]. Selecting the optimal kernel function frequently requires experimentation with a set of well-known functions, and occasionally hand-crafted kernel functions are used, which is a laborious process and does not ensure that the optimal option was tested [15]. In regression, SVM seeks to identify a function that maintains an ϵ -tolerance of the target values while minimizing the norm of w in order to maintain the model as flat as possible [18]. Even though SVMs often do well and achieve excellent accuracy in high-dimensional environments, their performance suffers when dealing with datasets

that contain noisy or overlapping classes [16].

Naïve Bayes

Two main principles underlie the supervised classification method known as Naive Bayes: the naive assumption that all features are conditionally independent of one another and the application of Bayes' theorem, which Thomas Bayes first identified in the 18th century [11]. Bayes' theorem states that:

$$E1 : \quad P(A|B) = \frac{P(B|A) P(A)}{P(B)} \quad (2.1)$$

where A and B are events, $P(A)$ and $P(B)$ are their probabilities, and $P(A|B)$ is the conditional probability of observing A given that B has occurred [11]. The theorem is reformulated for classification in ML as:

$$E2 : \quad P(Y|X_1, X_2, \dots, X_j) = \frac{P(X_1, X_2, \dots, X_j|Y) P(Y)}{P(X_1, X_2, \dots, X_j)} \quad (2.2)$$

where the posterior $P(Y|X_1, X_2, \dots, X_j)$ is the probability of class Y given the feature values, the likelihood $P(X_1, X_2, \dots, X_j|Y)$ is the probability of the features given the class, $P(Y)$ is the prior probability of class Y before observing any features, and the marginal probability $P(X_1, X_2, \dots, X_j)$ is constant across classes, so classification is based on comparing posterior numerators [11]. Because it presumes independence between each pair of features and performs well for both binary and multi-class datasets, including spam filtering and text categorization, Naive Bayes is often utilized for classification applications [16]. Its parameters can be estimated with only a small amount of training data, it is resistant to noisy instances, and it is computationally efficient [16]. Gaussian, Multinomial, Complement, Bernoulli, and Categorical Naive Bayes are all typical variations [16]. Because of their minimal computational expense, Naive Bayesian classifiers are

frequently used in sentiment analysis, but their independence assumptions may lead to inaccurate conclusions [11]. Because Naive Bayes requires little iterative parameter estimation, it is simple to implement and practical for use with massive datasets [11].

K Nearest Neighbors

K-Nearest Neighbors is a simple and widely used supervised learning algorithm [11]. It is an instance-based, or "lazy learning" method that stores all training instances in n -dimensional space rather than constructing an explicit global model [16]. Distance functions such as Euclidean, Manhattan, Minkowski, or Hamming distance are commonly used to classify new data points based on similarity measures, particularly for categorical variables [18]. A new object's predicted class is determined by a majority vote among its k nearest neighbors, where the object is assigned to the class that is most frequent among those neighbors [18]. The accuracy of KNN depends heavily on data quality, but it performs well for multi-class classification tasks and is relatively robust to noisy training data [16]. The dataset is simply stored during the training phase, requiring minimal computation; however, the testing phase involves significant computation due to distance calculations required to identify the k nearest neighbors for each test instance [13].

The Minkowski distance, computed between two points $U = (U_1, U_2, \dots, U_n)$ and $V = (V_1, V_2, \dots, V_n)$, is given by:

$$E3 : \quad d_q(U, V) = \left(\sum_{i=1}^n |U_i - V_i|^q \right)^{\frac{1}{q}} \quad (2.3)$$

which generalizes both the Euclidean distance (when $q = 2$) and the Manhattan distance (when $q = 1$) [11]. While Euclidean distance represents the straight-line

distance between two points, Manhattan distance represents the distance a car would travel in a grid-like city structure [11]. The main challenge in applying KNN for both classification and regression tasks is selecting an appropriate value for k [16]. K Nearest Neighbors is frequently used to determine the class of new data points, and as illustrated in Figure 1, the predicted class is typically the one receiving the majority vote among the k closest neighbors [11].

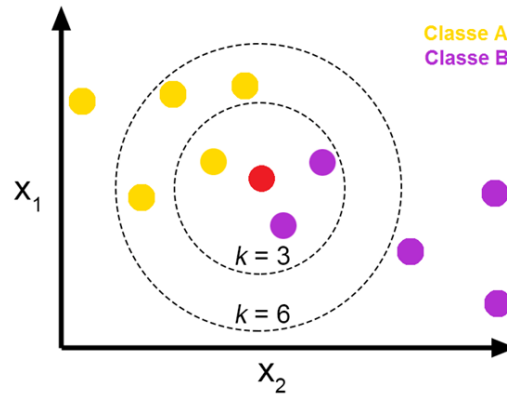


Figure 2.1: K-Nearest Neighbours Algorithm [11]

Artificial Neural Networks

Computational models based on the structure and operation of the human brain are known as neural networks. Each neuron serves as a basic processing unit, and neurons are connected via weighted links that transmit signals. These neurons are arranged in layers known as input, hidden, and output. The activation of a neuron is usually a number between 0 and 1, which is calculated by the inputs it receives and sends to the next layer using a transfer function [11]. Although neural networks are capable of performing regression and classification tasks, the majority of networks are built to only handle one task at a time. Multiple output units may

be necessary to perform classification across many states, and the outputs may need to be converted into variables during a post-processing step. Cross-talk is possible in networks with several outputs, which can make it challenging for hidden neurons to learn more than one function at once. One popular solution is to train distinct networks for each output and then combine them into an ensemble [17].

Artificial neural networks (ANNs) are a set of ML techniques that are modeled after biological neural networks. The Multi-Layer Perceptron, which Frank Rosenblatt invented in the late 1950s, is thought to be the first neural network that was ever completely functional. It is commonly used in image identification and data categorization since it employs gradient backpropagation for training [11]. One of the main benefits of ANNs is that they can replicate nonlinear relationships and dynamic interactions without requiring specific user input [15]. A significant drawback, though, is that the model's internal processes are frequently hard to understand. Understanding the connections between input and output layers is difficult, especially in deep ANNs with several hidden layers, because hidden layers primarily mediate effects. Using intuition or causal reasoning, it is difficult to comprehend the impact of specific variables or connection weights due to this opacity, despite efforts to make it more understandable. Furthermore, there are several variables in the training of ANNs that need technical knowledge to set up and improve. Large or deep networks may also need a large number of samples, sometimes in the tens or hundreds of thousands, in order to perform at their best, which might be expensive for some uses [15].

K-Means Clustering

The well-known unsupervised learning technique known as K-means clustering

divides a dataset into k separate clusters depending on feature similarity [17]. First, k initial centroids are selected either randomly or via a heuristic, and then, in an iterative process, each data point is assigned to the nearest centroid [16]. The centroids are then recalculated as the average of all the points within each cluster. Convergence is achieved when the centroids stop moving, and this process continues [17]. Figure 2 depicts a K-means flow chart.

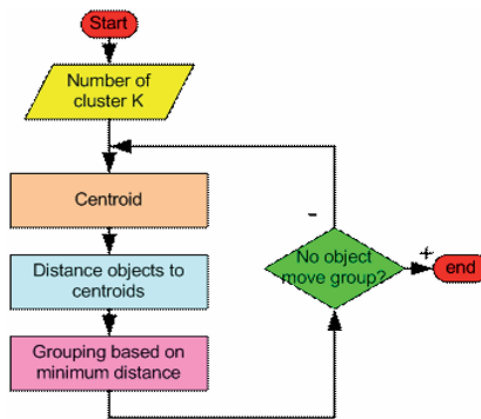


Figure 2.2: K-means iteration [17]

The Iterative Process:

1. Find the center coordinate.
2. Ascertain the distance between each item and the center.
3. Use the minimum distance to group the item.

K-means attempts to reduce the sum of squared distances between data points and their corresponding cluster centroids by effectively reducing intra-cluster variance [17]. K-means is susceptible to outliers and extreme values, even if it is quick, simple, and useful for well-separated datasets, since outliers and extreme values can have a big impact on the mean. K-medoids and other modifications increase

resilience to noise and outliers [16]. The k-means clustering method is widely recognized for its simplicity, ease of understanding, and computational efficiency [15]

One of the limitations of K-means is that it depends on the initial placement of centroids, which can lead to inconsistent results and subpar clustering [17, 16]. In addition, the need to define the number of clusters in advance is a severe constraint. Not all methods provide consistent estimations of k , and even a small shift in k can result in significant changes. The algorithm also employs Euclidean metrics to determine the distance between observations and cluster centroids, which assumes that the within-cluster variance is the same for all clusters. It can produce counter-intuitive results when some clusters are significantly larger than others [15]. With varying initializations, the algorithm might need to be run numerous times in order to produce more reliable clusters. Despite these challenges, K-means remains a fundamental and adaptable technique in data analysis and ML applications, such as image-based identification and feature grouping tasks [16].

2.3 Face Recognition Using Machine Learning

The act of identifying a person's face using a vision system is known as FR. It has been a vital instrument for human-computer interaction because it is used in business environments, security systems, and video monitoring. It is also used on social media platforms such as Facebook. Face recognition has gained popularity since the rapid growth of artificial intelligence because of its intrusive character and its position as the primary means of human identification in comparison to other biometric techniques. Without the subject individual being aware of it, it

may be readily assessed in an unregulated setting [3].

The majority of record defeats achieved by DL have been in semantic segmentation, image recognition, natural language processing, and a variety of other practical applications. Different DL methods are used in face identification, including Stacked Autoencoder, Convolution Neural Network (CNN), and Deep Belief Network (DBN). Convolution Neural Network is often used as an algorithm in face and image recognition. In order to augment the number of features, CNN is a type of artificial neural network that employs a convolution technique to extract features from the input data [3].

2.3.1 Machine Learning Techniques Applied to Face Recognition

Algorithms differ in how they extract and represent facial features. With minimal computing effort, classical techniques like Principal Component Analysis (PCA) and Local Binary Patterns (LBP) prioritize dimensionality reduction and texture-based features, whereas sophisticated methods like SVMs and CNNs employ high-dimensional feature spaces and hierarchical learning to attain superior accuracy, particularly in difficult scenarios. The main algorithms are discussed in this chapter [4].

PCA-Based

A statistical technique called PCA employs orthogonal transformation to transform observations of possibly correlated variables into a collection of linearly uncorrelated variables known as principal components. It accomplishes this by projecting

the data into a new coordinate system where each principal component's variance is increased [4]

Principal Component Analysis, which is used in FR, identifies the major components of a collection of images and represents each face in a lower-dimensional space using eigenvectors linked to higher eigenvalues. Principal Component Analysis greatly improves the speed of recognition by reducing the picture size and focusing just on the most crucial elements [19]. "EigenFaces" for identification are generated by the procedure, which involves calculating the covariance matrix, difference vectors, average image, and eigenvectors [19].

Principal Component Analysis has benefits including data visualization, feature transformation, dimensionality reduction, and improved ML performance. Additionally, it has greater FR accuracy in cases where changes in expression or background affect the face. Due to these advantages, PCA is frequently used in FR applications in conjunction with other algorithms [4].

SVM-Based

Due to their ability to tackle linear and nonlinear classification tasks with minimal computing power, SVMs are a popular choice for FR applications [4]. Support Vector Machines are well suited for FR applications that are non-linearly separable and have high dimensions because they operate by identifying the most discriminative information in the training data and determining the best hyperplane to separate different classes [20, 12]. Support Vector Machines can be used for FR either on the original image data or after features have been extracted from the face.

They can also be coupled with other techniques to increase performance [12]. Under rigorous validation methods, multi-scale facial features combined with SVMs attain high accuracy, and hybrid methods that incorporate additional feature sets with SVMs enhance performance in imbalanced datasets [4]. On complex datasets, more sophisticated SVM variants such as kernel SVMs, Directed Acyclic Graph SVMs, and Learning Distance Metric based models have attained higher recognition rates, reaching up to 99.85% [4]. Other variants, such as Least Squares SVM (LS-SVM) and component-based SVM classifiers, also offer quick computation and high accuracy, making them useful for multi-class FR tasks [12, 20]. In general, SVMs are still among the most popular and trustworthy methods for face identification because of their adaptability, excellent performance, and compatibility with various feature extraction techniques.

CNN-Based

Convolutional Neural Networks are a type of feed-forward, multi-layered neural network that has been shown to be very successful at FR and classification tasks [4, 3]. Unlike conventional ML algorithms that need manual feature extraction, CNNs are particularly well suited to analyzing visual data because they automatically learn and extract features from images. An average CNN architecture consists of convolutional layers, pooling layers, rectified linear unit activations, and fully connected layers. This architecture employs convolutional layers to recognize facial characteristics, pooling layers to decrease dimensionality, and fully connected layers to do classification [3, 11].

The performance has been improved by the development of several CNN architectures. The first CNN, LeNet, was created for low-resolution grayscale photos,

but more complex architectures like AlexNet and ResNet handle higher-resolution images and address issues like the vanishing gradient problem through stacked convolutional layers and skip connections, respectively [11]. By linking each layer to all following layers, DensNet further enhances training by giving direct access to gradients and feature maps [11].

Changes in lighting, facial expressions, occlusions, pose variations, and low-resolution pictures are just a few of the major variables that CNNs are resistant to. For example, when performing identification in under 0.01 seconds, a 4-layer CNN attained 99.5% accuracy on the AR dataset and 85.13% on the FERET dataset [4]. The accuracy of coupled architectures for high- and low-resolution images increased to 97.2% even for extremely low-resolution images of 6 by 6 pixels [4]. Additionally, CNNs have been combined with dimensionality reduction techniques like PCA and similarity scoring frameworks like joint Bayesian frameworks, achieving 98.52% accuracy on the CAS-PEAL dataset [4]. Furthermore, CNNs have proven their versatility, speed, and excellent recognition rates in niche applications like age estimation and secure face verification [4]. CNNs are typically a robust and adaptable solution for identifying facial features in challenging situations and large datasets.

LBP-Based

The texture-based operator known as LBP assigns labels to pixels by thresholding the neighborhood around each pixel and converting the result into a binary number. This method successfully captures local characteristics in images. It is One of the easiest yet powerful face identification algorithms, which can enhance detection accuracy when used in conjunction with methods like the Histogram

of Oriented Gradients [4]. It works well in regulated settings and is resistant to monotonic grayscale changes.

Additionally, researchers have improved performance by combining LBP with additional preprocessing and classification techniques. For instance, the discrete wavelet transform has been employed to minimize storage needs and extract key features, attaining 98.28% accuracy on noisy images and 99.3% accuracy on noise-free images on the MIT face dataset [4]. In general, methods based on LBP are effective, adaptable, and computationally straightforward for face identification, particularly when used in conjunction with additional feature extraction or categorization strategies.

2.3.2 Datasets Used in Face Recognition

Face database is a collection of face images that can be used to test and evaluate the performance of FR systems. The database used to create algorithms is reflected in their outcomes. The databases most frequently used to assess the algorithm's performance are listed below [13].

AR Database

The AR database is a publicly available face image dataset containing between 3,000 and 4,000 color images of 126 subjects, including 70 males and 56 females, taken in frontal view under regulated settings [20, 13]. The photos exhibit changes in facial expression, lighting, and obstructions such as sunglasses, hairstyles, and scarves [20, 13]. Temporal variations were represented in the dataset because each subject was photographed during two distinct sessions that occurred 14 days apart

[13]. The AR database is widely used in FR research and is freely available for academic use [20].

The ORL Database

The ORL database consists of 400 grayscale images from 40 subjects, each of whom provided 10 images shot between 1992 and 1994 at the AT&T Cambridge Laboratory in England [11]. The photographs, taken against a dark, uniform background, capture differences in lighting, facial expressions (smiling or not smiling, open or closed eyes), and facial characteristics (wearing glasses or not) [11, 19]. The subjects are photographed in an upright frontal position, allowing for minor side movement of up to around 20 degrees [11, 19]. The database is frequently used for comparing FR algorithms since each picture is 92 by 112 pixels in size with 256 grayscale levels [19].

Yale Face Database

The Yale Face Database [11] comprises 165 grayscale GIF images of 15 people, each with 11 images showing different facial expressions and configurations, such as center light, left light, right light, with glasses, without glasses, happy, sad, sleepy, normal, surprised, and wink. The database was produced under complete supervision at Yale University's Center for Computational Vision and Control [11]. It comes in two volumes. The Extended Yale Face Database B consists of 2,414 photos of 38 participants, with an emphasis on changes in lighting and without any changes in facial expressions or occlusions. In contrast, Yale Face A, also known as yalefaces, has 15 subjects, including 14 males and one female, with varying expressions and lighting conditions, such as sad, normal, and happy, as well as

images with and without glasses [20]. For feature extraction applications, it is offered in a cropped form [20].

LFW Database

The big face image database called Labeled Faces in the Wild (LFW) was developed to aid in study on unconstrained FR. Using the Viola-Jones face detector [11], which identifies and centers faces in photos taken from the internet, it includes more than 13,000 face images. Approximately 1,680 of the 5,749 people in the dataset are seen in two or more photographs [11]. The photographs are available in four versions at LFW: the original set and three aligned variations. Among these are the "funneled" images known as LFW-a and the "deep funneled" variants, both of which were created to enhance face verification capabilities [20].

FERET Database

The FERET database, a well-known facial recognition dataset obtained between 1993 and 1996 under the FERET program a partnership between DARPA and NIST contains frontal face images of more than a thousand people, along with metadata such as facial points, ethnicity, date of birth, gender, and illumination conditions [13, 20]. The high-resolution 24-bit color version released by DARPA in 2003 included 2,413 photos depicting 856 people [20]. The main purpose of FERET was to offer a uniform dataset for assessing FR algorithms and systems, as well as to aid in their creation. Because the database is heavily biased towards young, Caucasian participants, its use in age or race-based analyses is restricted, even if it has been used in studies on gender categorization [13].

2.3.3 Evaluation Criteria

Metrics like accuracy reflect the overall correctness of the classification [7] but, accuracy alone is insufficient for assessing a classifier’s performance since it only measures the percentage of correctly classified instances and ignores other critical variables. The quantity of positive results in comparison to all expected positives, the accurate distribution of instances across all classes, and the correlations between data characteristics are some of the variables [18]. For a thorough evaluation, more factors are thus required. The confusion matrix, which is an n by n table where the rows reflect the actual class labels and the columns represent the expected classes, is a key instrument for this assessment.

		Predicted	
		0	1
Actual	0	TN	FP
	1	FN	TP

Figure 2.3: Confusion matrix [18]

From the confusion matrix, the key metrics are as follows: True Negative (TN): the proportion of negative cases correctly classified, calculated as $TN / (TN + FP)$. False Positive (FP): the proportion of negative cases incorrectly classified as positive, calculated as $FP / (TN + FP)$. False Negative (FN): the proportion of positive cases incorrectly classified as negative, calculated as $FN / (FN + TP)$. True Positive / Recall (TP): the proportion of positive cases correctly classified, calculated as $TP / (FN + TP)$. Accuracy: the overall proportion of correct predictions, calculated as $(TN + TP) / (TN + FP + FN + TP)$.

+ FP + FN + TP) [18].

precision, recall, false acceptance rate (FAR), and false rejection rate (FRR) are other metrics that can be used to evaluate performance. Recall calculates how many true matches are retrieved, precision quantifies how many predicted matches are right. The FRR and FAR are frequently used in real-world biometric systems to evaluate performance [7]. The inverse relationship between rejecting imposters and incorrectly rejecting authorized users is reflected in the trade-off between FRR, also known as Type I Error, and FAR, also known as Type II Error. Designers can strike a balance between these rates in accordance with the demands of the application thanks to adjustable thresholds. A system that has both a low FAR and a low FRR has a high degree of discrimination. A helpful indicator of overall performance is minimizing the area under the FAR-FRR curve, especially where the FAR and FRR intersect. To accomplish this, it is necessary to choose the right sensors, feature extraction techniques, and pattern matchers so that the classifier functions consistently across a variety of settings [7].

2.4 Summary

The basic principles of FR and its combination with ML to improve performance in real-world situations are covered in this chapter. With applications in surveillance, access control, security, and pervasive computing, FR is a popular biometric technique for identifying people by analyzing their facial characteristics. Age, gender, race, image resolution, occlusion, lighting, and pose are just a few of the variables that affect its efficiency.

The performance of FR systems depends heavily on ML, which includes supervised, unsupervised, semi-supervised, and reinforcement learning. K-means clustering, Naive Bayes, SVM, ANN, and KNN are some of the popular algorithms used in FR. Every method has its advantages and disadvantages.

Face Recognition utilizes strategies like PCA for dimensionality reduction and feature extraction, as well as SVM for classification. Face Recognition models are trained and assessed using datasets such as LFW, ORL, Yale, and FERET. Performance is influenced by covariates such as posture, lighting, occlusion, and age. Accuracy and resilience are assessed using evaluation criteria, which directs advances in FR technology.

Chapter 3

Research Methodology

A structured research methodology is essential to ensure that the proposed solution is both valid and reliable. In this thesis, a systematic approach is adopted to investigate FR under uncontrolled conditions.

3.1 Introduction

This chapter outlines the methodology used to conduct the research presented in this thesis. It describes the identified research gap, the formulation of the problem statement, and the research questions and aims guiding the work. The chapter then explains the Design Science framework adopted (DSR), followed by an overview of the tools and software selected for implementation and experimentation. The chapter concludes with a summary of the methodological approach.

3.2 Research Gap

Face recognition has made great strides thanks to ML and DL, but existing systems continue to struggle in keeping high accuracy in unpredictable real-world settings. The majority of current models only function well on datasets collected in regulated settings where lighting, posture, facial expression, and occlusion are constant. But in real-world circumstances like surveillance film or images taken with mobile devices, faces frequently have poor resolution, illumination changes, non-frontal poses, blurring, and partial occlusion.

Although numerous studies have concentrated on enhancing specific aspects like feature extraction or classification, ML methods still lack a cohesive, reliable framework that can consistently address these integrated issues. The current techniques tend to deteriorate noticeably when several covariates occur at the same time. As a result, a more trustworthy FR model is needed that can perform consistently in actual, uncontrolled situations.

3.3 Problem Statement

Because of changes in lighting, pose, occlusion, and image resolution, current FR systems have difficulty performing consistently in uncontrolled settings. Due to these inconsistencies, the quality of feature extraction and the accuracy of recognition are both diminished. The issue addressed in this research is the absence of a strong FR model that can perform well in such real-world scenarios.

3.4 Research Questions and Aims

Research Questions:

- In uncontrolled settings, how can FR accuracy be increased?
- What preprocessing or feature extraction methods are available to improve recognition accuracy in situations such as low resolution, pose variation, and occlusion?
- How can ML or DL methods be combined to produce a more reliable FR system?

Aims:

- To compare the effectiveness of various preprocessing techniques in reducing covariate effects.
- To compare the effectiveness of different ML techniques for reliable recognition.
- To propose an improved approach that outperforms conventional methods on real-world images.

3.5 Design Science

The study of man-made items and engineering are the foundations of the Design Science Research (DSR) methodology. In essence, it offers a methodical approach to problem-solving by advancing human understanding through the invention of new tools and the development of design knowledge based on realistic solutions to actual problems [21].

3.5.1 Definition

The goal of design science is to develop and evaluate IT-based solutions to particular issues facing organizations. This entails creating the artifact, analyzing its effectiveness, adding new insights to current research data, and sharing the findings with the intended audience. A variety of forms are possible for artifacts, such as models, methodologies, prototypes, social advancements, or novel features of informational, social, or technical resources. An artifact is, broadly speaking, anything created to address a known research issue [22]. The Design Science Research Methodology (DSRM) process model, which is shown in Figure 3, emphasizes the crucial steps involved in designing, developing, and testing an artifact.

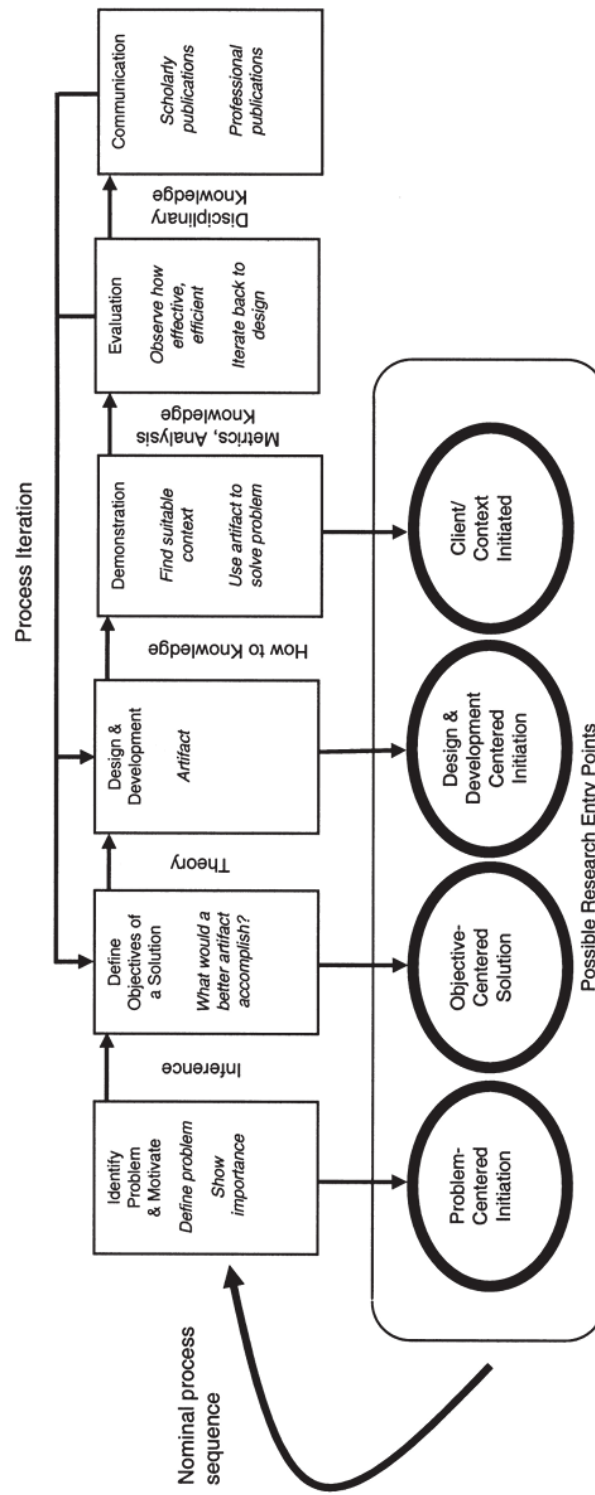


Figure 3.1: DSRM process model [22]

According to DSR guidelines, the study should result in an artifact that was purposefully designed to address a pressing issue that has not yet been resolved. Its usefulness, quality, and efficacy must be carefully assessed. Rigorous processes must be followed throughout the development and evaluation of the artifact, and the study must make a verifiable contribution [22].

3.5.2 Explanation

Describing an IT system at any level of detail, with an emphasis on its structure, processes, or component interactions, is referred to as system design. Examples include business process maps, database layouts, enterprise architecture, and software architecture. Requirements outline what a system must accomplish, such as "A life-support system must never fail" or "A multi-user system should protect private data from other users." These regulations assist in focusing and restricting the available design options [23].

A method is a series of steps that aid in the development of a system and typically specifies anticipated results and duties. Requirements analysis methodologies, enterprise architecture strategies, and software engineering methodologies are examples. An algorithm also outlines a sequence of actions, but it is carried out by a computer and serves as a specification for system behavior that may operate autonomously. Data analysis algorithms, communication protocols, and sorting methods are some examples [23].

Patterns offer shared design ideas that may be applied to a variety of systems. They provide assistance with the design process without going into detail about system-specific solutions. Service-based architectures, asynchronous messaging,

and singleton patterns are a few examples. "Do not add more personnel to a late project" or "Use pair programming to maintain high code quality" are two examples of the broad development recommendations provided in the guidelines, but the specific design itself is not discussed [23].

Ideas and connections that aid in system development are provided by languages and notations. Object-oriented programming languages, company process modeling notations, and entity relationship diagrams are a few examples. Metrics are frameworks used to evaluate the individual parts of a system or its architecture by providing numerical or descriptive assessments. Examples include business case analyses, architecture assessment tools, and measures such as cyclomatic complexity [23].

3.5.3 Examples

Consider a project aimed at developing a novel electronic health record system as an example of the needs of design science. In order to be considered design science, this project must satisfy three fundamental requirements. First, it necessitates a well-defined strategy for conducting research on the issue and obtaining data from stakeholders. This entails choosing the right data gathering methods, such as surveying healthcare workers and interviewing doctors or managers, as well as using the right ways to analyze and assess the system that has been developed [24].

Second, the project's findings must be related to current understanding in health informatics and information systems. This includes mentioning well-known theories, models, and related artifacts, such as other EHR systems. The distinctness

and correctness of the new solution may be properly evaluated by comparing the results to existing knowledge [24].

Third, the project’s findings must be related to current understanding in health informatics and information systems. This includes mentioning well-known theories, models, and related artifacts, such as other EHR systems. The distinctness and correctness of the new solution may be properly evaluated by comparing the results to existing knowledge [24].

Third, the project should successfully disseminate its findings to both academic and industry stakeholders. This may involve speaking at health-related events, going to conferences, and writing articles [24].

A partnership between university scholars and publishing firms is another example of design science. The initiative intends to create a cutting-edge tool that enhances the reading experience using internet and mobile technology as a way to address the problem of falling book sales and declining reading interest, particularly among young people. It is demonstrated in this example how design science can be used to solve real-world issues by generating and assessing novel solutions [24].

3.5.4 How Design Science Connects to this Research Work

Because the work centers around creating and evaluating a novel artifact to solve a real-world problem, DSR is directly applicable to this thesis. The artifact is an improved FR system that makes use of ML approaches. The thesis uses the DSR methodology to build a system that deals with real-world issues encountered in uncontrolled settings, such as shifting lighting, pose variations, occlusion, and

poor image resolution.

The thesis, in accordance with DSR principles, seeks to create scientific knowledge in addition to a practical solution. This will be accomplished by thoroughly documenting the design decisions, outlining the system's development, analyzing its performance, and sharing lessons learned throughout the process. By doing this, the research should add novel insights about enhancing face identification in challenging situations in addition to a functional artifact.

3.6 Possible Tools/Software Packages to Utilize

Python

Python is a high-level programming language that is easy to read and use, designed to be object-oriented and dynamically typed [25]. Its straightforward syntax and the wide range of external libraries like NumPy, OpenCV, Dlib, OpenFace, SciPy, and Matplotlib make it a great choice for activities such as web development, data analysis, artificial intelligence, and tasks in computer vision like processing images and recognizing faces [26]. Although Python runs slower than languages like C and C++ because it is interpreted, it enables quick development and is simple to learn, which makes it easy for software developers to pick up [26]. Its flexibility and capability have led to its widespread use in education, business, and significant projects, including managing systems at NASA, creating special effects at Industrial Light and Magic, running discussion forums at Yahoo! , and powering parts of Google's search engine and web crawler [26].

Java

Java is an object-oriented programming language that can be used on any platform, making it a must-have for software development [25]. It has been chosen for several studies on FR due to its trustworthiness, safety, and ability to work well on different operating systems [27]. A key benefit of Java is its built-in garbage collection system, which helps manage memory by freeing up space from unused items and reducing the chances of memory issues. The language also allows for creating modular and reusable code through its object-oriented approach, which is ideal for developing complex systems like FR programs. Furthermore, Java has a wide range of open-source libraries and APIs that help include image processing and ML tools, making the development process easier. On the downside, Java's lengthy code and slower performance compared to lower-level programming languages can be a drawback in projects that need quick processing or high responsiveness [25].

C++/.NET

Microsoft launched the .NET framework at the same time as the C# programming language, creating a system that covers all parts of software development, from the core operating system and up. The .NET framework offers a single application programming interface (API) that combines different technologies for displaying information, components, and data, including COM+ services, XML protocols, and web service standards such as SOAP, WSDL, and UDDI. C# is the main language used with .NET, created to provide a good mix of being easy to use, expressive, and efficient. It takes features from C, C++, and Java, ensuring type safety, supporting object-oriented design, and allowing it to run on different platforms. These traits make it a good choice for effectively building ANNs and other models in ML. Using object-oriented programming, we can represent parts

of the neural network as separate classes, which supports flexible design and quick testing of ideas. Various C# implementations for ANNs are available, such as the C# Neural Network Package and Neuro. NET version 2. 0, but some of these may need a lot of computing power and time for training [28].

Wolfram Mathematica

Mathematica is a widely used, potent computational software program that offers an environment for carrying out symbolic and numerical calculations, producing visualizations, and solving difficult issues in mathematics, science, and engineering. It enables users to carry out arithmetic operations, manipulate mathematical expressions, and solve equations by utilizing a large library of pre-built functions for activities like numerical computations, algebra, calculus, and linear algebra. In Mathematica, variables hold values that may be accessed during calculations, whereas functions include a set of instructions that can be used repeatedly with various arguments. Lists are a fundamental data structure that allows you to store collections of values and use built-in methods to add, remove, sort, and retrieve items. Using functions like Plot, ParametricPlot, Plot3D, and ParametricPlot3D, Mathematica also provides robust 2D and 3D graphing capabilities that allow users to modify axes, ranges, colors, and styles. Decision-making, iteration, and data collection operations are made possible by control structures such as conditionals (If, Switch), loops (For, While), and functional programming constructs (Map, Fold, Nest). Lastly, modules and blocks offer methods for developing modular and reusable code with local variables that are restricted to its scope, resulting in improved structure and less possible conflicts [29].

Mathworks Matlab

MATLAB is an interactive computer program that serves as a convenient “laboratory” for computations involving matrices. It can do basic tasks like solving equations at the same time and flipping matrices, as well as handling both symmetric and nonsymmetric eigenvalue problems. It even has advanced matrix functions like singular value decomposition [30].

MATLAB is a strong tool for numerical computing that can be used to create a face detection system. The system uses MATLAB functions, libraries, and toolkits for steps like preparing the data, extracting important features, and recognizing patterns. It is made to analyze images or video clips instantly, allowing for quick detection of faces in live video or saved media. Additionally, MATLAB has great visualization features that can show the detected faces along with their bounding boxes for clear visual results [31].

Choice of Language/Program

Python was chosen for this thesis because it is ideal for creating ML and computer vision projects, as mentioned earlier. Its straightforward language and adaptability made it easy to set up the suggested FR system, and its large collection of libraries helped with important functions like image processing, feature extraction, and model evaluation. These features allowed for quick development and testing, making Python a sensible and efficient option for building and testing the system within the scope of this project [26].

3.7 Summary

The approach for building a trustworthy FR system for real-world situations is outlined in this chapter. Problems with current FR systems include lighting changes, pose fluctuations, occlusion, and low-resolution images, even if they function well in regulated environments. The thesis adheres to the DSR approach, which emphasizes the creation, testing, and documentation of an artifact as a solution to a real-world issue. The creation of cutting-edge reading instruments or electronic health record systems are two examples of DSR applications.

Different programming languages and software are taken into consideration, such as Mathematica, MATLAB, Python, Java, and C#/.NET. Because it combines symbolic and numerical computation, data visualization, and ML capabilities, Python is chosen for its ability to effectively develop and assess the FR system.

Chapter 4

Face Recognition using Machine Learning

This chapter presents the application of the DSRM to the development and evaluation of the proposed FR system. Each stage of the DSRM process is discussed in relation to the implemented solution, demonstrating how the research problem was addressed through systematic design, development, and validation. The chapter also analyzes the system's performance, highlights key findings, and discusses the limitations encountered.

4.1 The Design Science Research Methodology in Practice

The DSRM offers an organized approach for creating and assessing tools that tackle recognized issues. In this research, DSRM is used to direct the creation and use of a facial recognition system that can function effectively in unpredictable

settings. The next sections explain each step of the DSRM process and how it was applied in this thesis.

4.1.1 Activity 1: Problem Identification & Motivation

Face recognition systems have achieved high accuracy under controlled conditions; however, their performance significantly degrades in real-world environments due to various covariates such as pose variation, occlusion, and appearance changes. These challenges limit the reliability of such systems in practical applications, where input images are often captured under uncontrolled conditions.

The motivation for this research arises from the need to improve the robustness of FR systems when exposed to these variations. In particular, the inability of conventional approaches to generalize across changes in facial appearance highlights the necessity for enhanced preprocessing, feature representation, and classification strategies. Addressing these challenges is essential for developing systems that can operate reliably in real-world scenarios, such as biometric verification and user authentication.

4.1.2 Activity 2: Objectives of Solution

Based on the identified problem, the objective of the proposed solution is to develop a FR system that is robust to common real-world variations. The system is designed to improve recognition performance in the presence of challenges such as pose variation, minor occlusions, and appearance changes.

To achieve this, the solution incorporates data augmentation techniques to simulate realistic conditions, utilizes DL-based feature extraction to generate distinc-

tive facial representations, and applies ML models for reliable classification. In addition, the system integrates multiple decision-making strategies to enhance prediction reliability and reduce the likelihood of misclassification.

In summary, the aim is to enhance classification effectiveness in challenging scenarios by increasing durability and providing more reliable recognition outcomes in uncontrolled environments. Additionally, the success of the proposed solution is assessed using performance measures like accuracy and classification dependability, making sure that the system shows significant progress compared to the baseline performance.

4.1.3 Activity 3: Design & Development

The design and development phase focuses on the creation of the proposed FR system as an artifact that addresses the identified challenges. The system was designed to operate under uncontrolled conditions by incorporating multiple components, including data preprocessing, feature extraction, classification, and decision-making.

The desired functionality of the proposed system is to accurately identify known individuals while reliably detecting unknown users under varying real-world conditions. The system is designed to process input images efficiently, extract meaningful facial representations, and make robust classification decisions even in the presence of moderate variations such as pose changes and minor occlusions. Additionally, the system supports user registration and dynamic retraining, allowing it to adapt and expand over time.

System Overview

The proposed system is designed as an end-to-end FR pipeline that takes an input image and determines whether the individual is a known or unknown user. From a user perspective, the system operates through a simple interface where an image is uploaded, after which the system automatically processes the image and returns a recognition result.

Internally, the system follows a structured sequence of processing steps. First, the uploaded image is processed to extract meaningful facial features using a DL-based model. These features are then transformed into numerical representations known as embeddings. The embeddings are normalized and passed to a classification model that predicts the identity of the individual. To improve reliability, an additional verification step is performed using similarity-based matching. Finally, the system combines the outputs of these components to produce the final decision.

The system architecture follows a modular pipeline design, where each component performs a specific function within the recognition process. This modularity allows for flexibility in improving or replacing individual components without affecting the overall system.

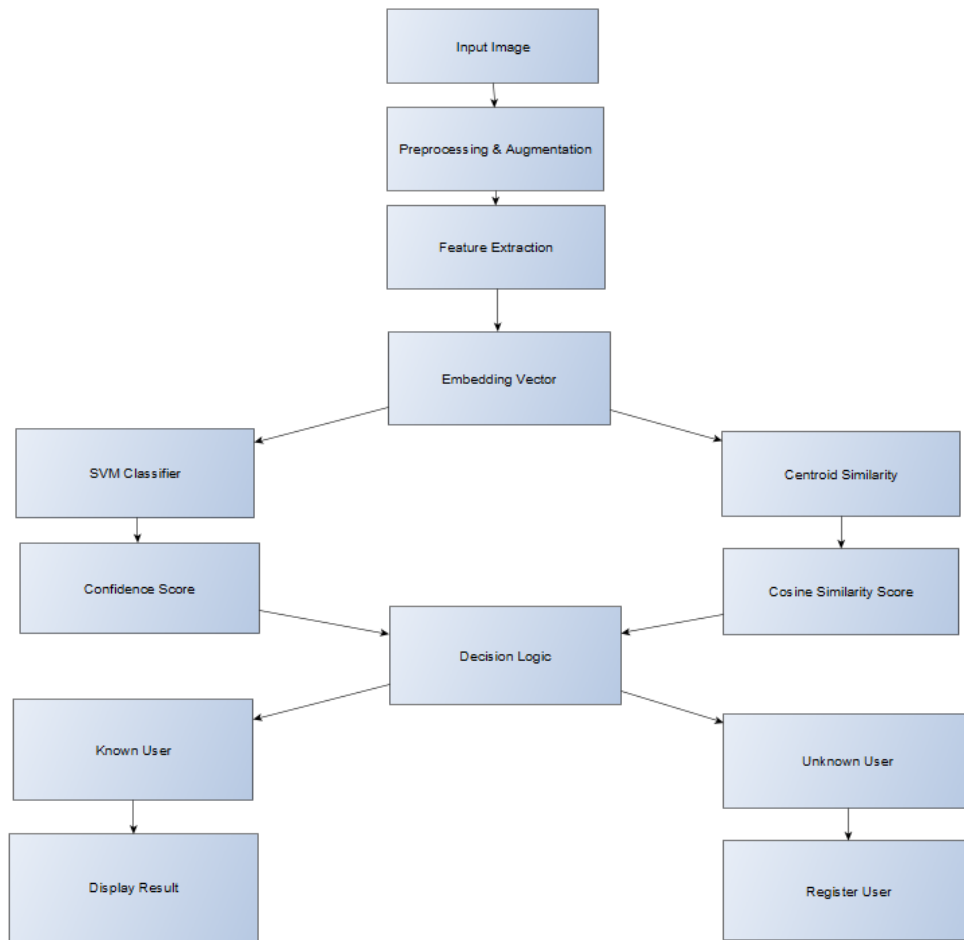


Figure 4.1: System Pipeline Diagram

Data Preparation and Augmentation

To improve the robustness of the system, data augmentation techniques were applied to simulate real-world variations that are not fully represented in the dataset. These transformations included adjustments to brightness, contrast, and color, as well as horizontal flipping and slight rotations of the images. The inclusion of rotation-based transformations was particularly important for simulating pose variation, which is a major factor affecting FR performance. By exposing the model to these variations during training, the system becomes more capable of generalizing to unseen conditions.

Feature Extraction

Feature extraction was performed using a DL-based model through the DeepFace framework, specifically utilizing the FaceNet model. This model converts facial images into high-dimensional numerical vectors, known as embeddings, which capture distinctive facial characteristics. To ensure consistency and improve comparison between embeddings, each generated vector was normalized. This step ensures that similarity calculations are not affected by scale differences and allows for more reliable matching between facial representations.

Classification Model

For the classification task, a SVM with a RBF kernel was employed. The SVM model was trained on the extracted embeddings to learn decision boundaries between different identities. The choice of SVM is motivated by its effectiveness in handling high-dimensional data and its ability to perform well in multi-class

classification scenarios. Additionally, probability estimates were used to provide a confidence score for each prediction, which is later used in the decision-making process.

Hybrid Decision Strategy

To improve the reliability of the system, a hybrid decision-making approach was used by combining the SVM classifier with a similarity-based method using class centroids. For each identity, a centroid vector was calculated by averaging all its embedding vectors. During prediction, the input embedding is compared to each centroid using cosine similarity to measure how closely it matches each known identity. The final decision is based on both methods. If the SVM and centroid-based approaches predict the same identity, that result is accepted. If they disagree, simple threshold rules are used to decide whether the input belongs to a known user or should be marked as unknown. This helps reduce incorrect or uncertain predictions that might come from relying on only one method.

System Interface and Interaction

The system was implemented using a Streamlit-based interface, allowing users to interact with the model in a simple and intuitive manner. Users can upload an image through the interface, after which the system processes the image and displays the recognition result along.

The interface also includes a sidebar that provides real-time system feedback and interaction history. This sidebar displays the current system status (e.g., ready, scanning, or training), allowing users to understand the state of the system during

operation. Additionally, a log of recent recognition activities is maintained, showing detected users and timestamps, which enhances transparency and usability. If the system identifies the user as known, the result is displayed and can optionally be stored in the database. If the user is classified as unknown, the interface allows for registration by entering a name, after which the system retraines to include the new identity. This interaction supports continuous system improvement and adaptability.

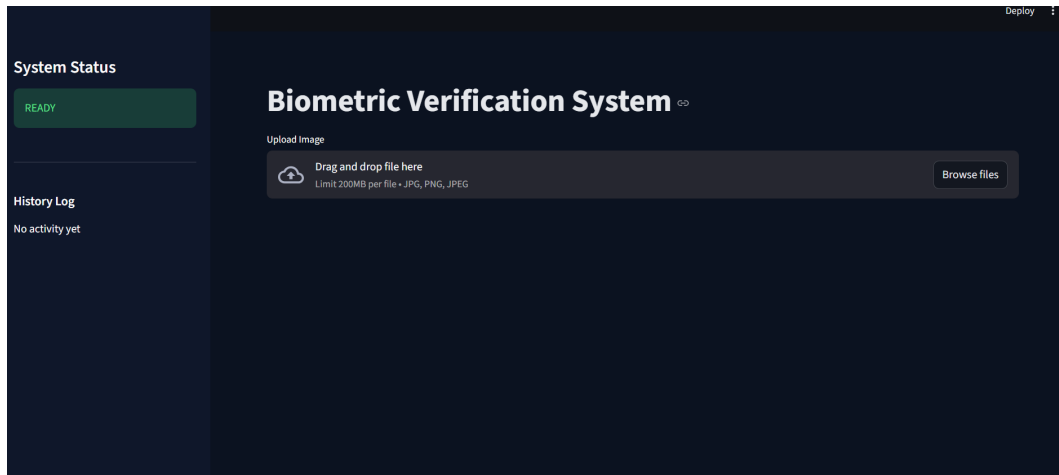


Figure 4.2: User Interface

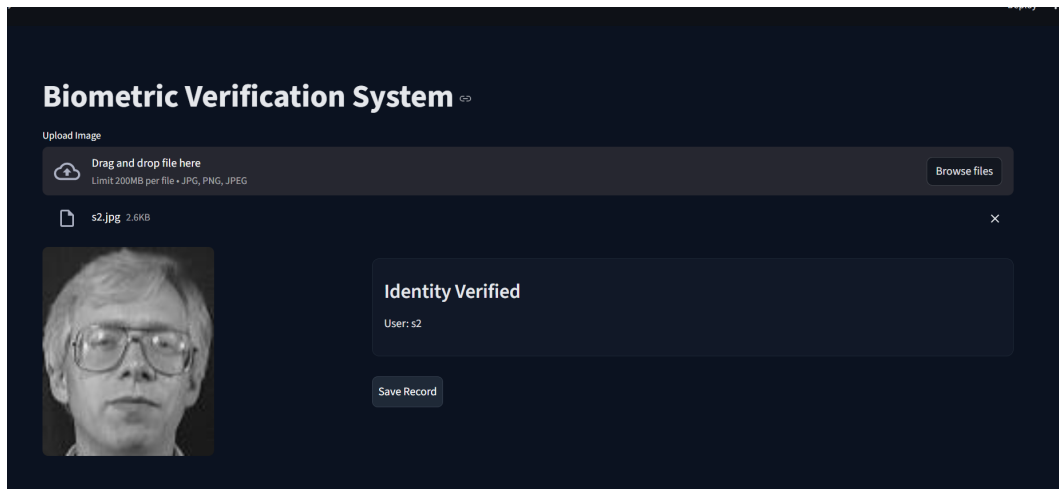


Figure 4.3: Known User Registration

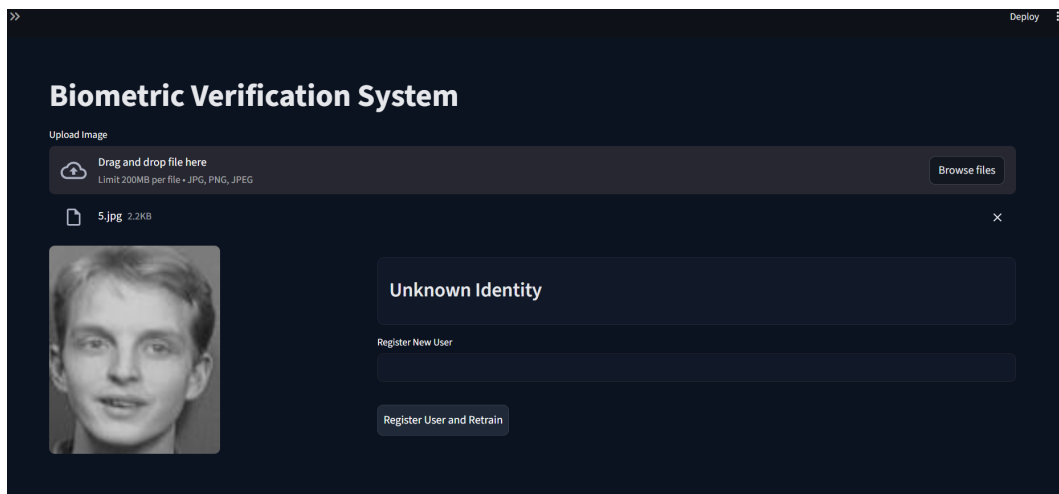


Figure 4.4: Unknown User Registration

4.1.4 Activity 4: Demonstration

The demonstration phase focuses on validating the functionality of the proposed system by applying it to different test scenarios. The system was tested using both known and unknown images to evaluate its ability to correctly identify individuals and handle unfamiliar inputs. To demonstrate its effectiveness, multiple types of test cases were conducted. First, images that were not included in the training set but belonged to known individuals were used to evaluate the system’s recognition accuracy. These images allowed for assessing how well the model generalizes to unseen samples of known users.

In addition, the system was tested using images of individuals that were completely excluded from the training dataset. These cases were used to evaluate the system’s ability to correctly classify inputs as unknown users, which is a critical requirement for real-world biometric systems.

To further challenge the system, additional test images were manually modified to simulate real-world variations. These modifications included changes in facial pose, including side-profile views, as well as the addition of visual elements such as glasses, facial hair, and minor occlusions, along with slight alterations in orientation. The system successfully demonstrated its ability to recognize known users under these variations and to correctly detect unknown individuals, indicating effective separation between known and unseen identities. The hybrid decision strategy contributed to improving reliability by reducing incorrect classifications, particularly in borderline cases where a single method alone might produce uncertain results.

Examples of the system's behavior under different scenarios are illustrated in the following figures, including known user recognition, unknown user detection, and performance under modified conditions.

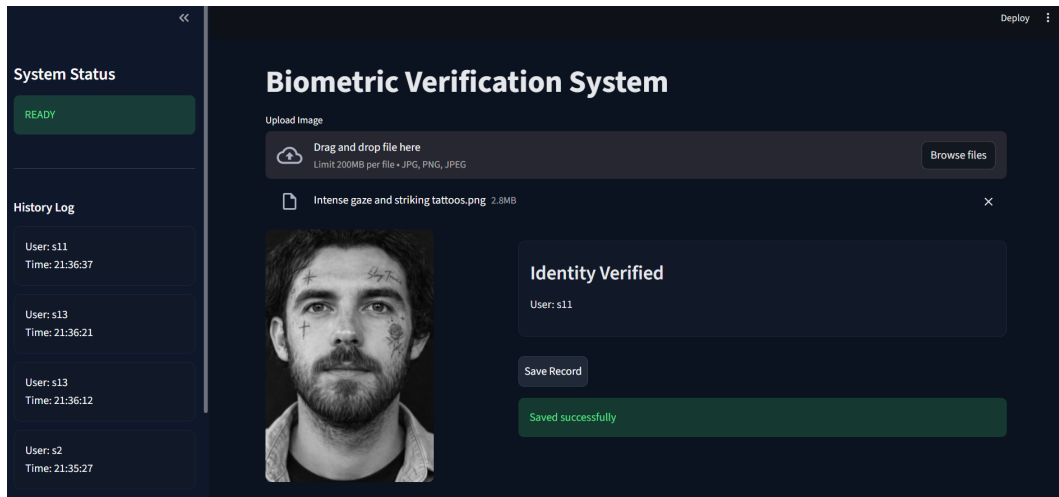


Figure 4.5: Recognition under Modified Conditions (Appearance Variation)

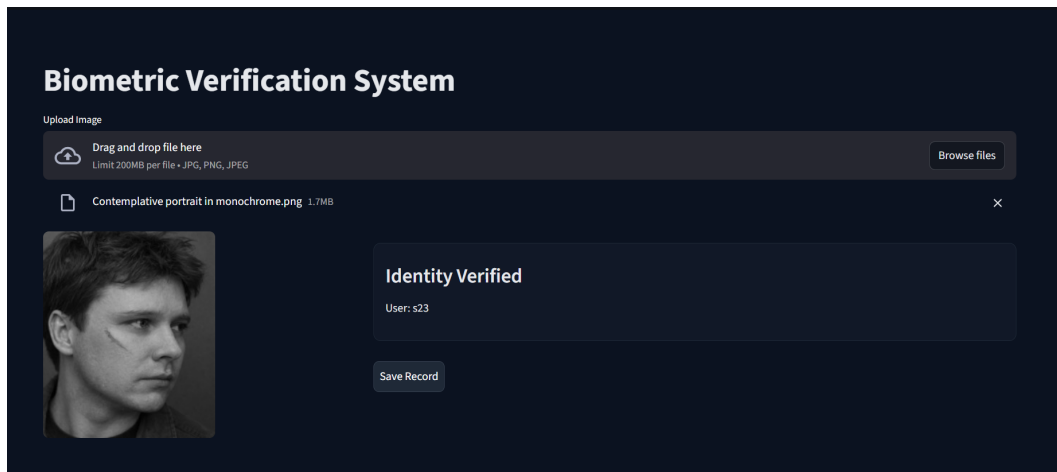


Figure 4.6: Recognition under Modified Conditions (Pose Variation)

4.1.5 Activity 5: Evaluation

The evaluation phase assesses how effectively the proposed system addresses the research problem by comparing the achieved results with the defined objectives. The performance of the system was evaluated using standard classification metrics derived from the confusion matrix, including accuracy, precision, recall, and F1-score.

The trained model achieved an overall accuracy of 1.00 on the test set, indicating that all test samples were correctly classified. In addition, precision, recall, and F1-score values for all classes were consistently equal to 1.00, demonstrating that the system successfully identified all known individuals without misclassification. These results indicate that both false positive and false negative rates are minimized under the evaluated conditions. The system was also evaluated on its ability to distinguish between known and unknown users. The results show that the system was able to correctly reject unfamiliar inputs while maintaining ac-

curate recognition of known individuals, confirming effective separation between known and unseen identities.

When compared to the defined objectives, the system demonstrates strong performance in handling variations such as pose changes and appearance modifications. This indicates that the proposed approach effectively improves reliability and maintains consistent performance under the tested conditions. While the achieved results are highly promising, they are based on the specific dataset and test scenarios used in this thesis. Therefore, further evaluation on more diverse datasets would be beneficial to fully assess the system's generalization capability.

4.1.6 Activity 6: Communication

The communication phase focuses on presenting the developed system and its results to relevant stakeholders and incorporating feedback to improve the overall design. In this thesis, the proposed system and its intermediate results were discussed with domain experts and academic supervisors throughout the development process. Feedback was obtained at multiple stages of the project, particularly after the initial system design and during the implementation phase. These discussions provided valuable insights into both the technical approach and the usability of the system, guiding several refinements to improve its effectiveness.

Specifically, experts emphasized the importance of ensuring reliable handling of unknown users, as this is a critical requirement in practical biometric systems. Based on this feedback, threshold-based decision-making was incorporated, and the hybrid strategy combining SVM classification with centroid-based similarity was introduced to reduce incorrect acceptance of unfamiliar identities.

Additionally, an important limitation identified during discussions was the system’s initial dependency on having multiple samples of a new user before achieving reliable recognition. This requirement was considered impractical for real-world applications, where systems are expected to operate with minimal user input. In response, the approach was refined to improve generalization from limited data by enhancing the embedding robustness and decision strategy, allowing the system to better handle new users with fewer samples. Furthermore, feedback highlighted the need to evaluate the system under more challenging conditions. As a result, additional testing was performed using manually modified images, including variations in pose, side-profile views, and appearance changes such as glasses, facial hair, and minor occlusions. This helped improve the robustness of the evaluation and provided a better understanding of the system’s behavior under non-ideal conditions.

Overall, these interactions contributed to iterative improvements in both the system design and its evaluation. The feedback process ensured that the developed solution aligns more closely with real-world expectations and user needs, while also strengthening the validity of the research outcomes.

4.2 Limitations

While the proposed FR system demonstrated strong performance under the evaluated conditions, several limitations should be acknowledged.

A primary limitation relates to the dataset used for training and evaluation. Although augmentation techniques were applied to simulate variations such as pose

changes and minor occlusions, the dataset remains relatively controlled. In particular, variations in lighting conditions, background complexity, and image quality were not extensively represented, which may affect performance in more unconstrained real-world environments. Another limitation concerns system scalability and adaptability. The current approach requires retraining the model whenever a new user is added, which can become time-consuming as the dataset grows and may limit real-time applicability in larger deployments.

Finally, the evaluation was conducted on a relatively limited number of subjects compared to large-scale FR systems. Although the results demonstrate strong performance, further validation on larger and more diverse datasets is necessary to fully assess scalability and generalization.

Chapter 5

Conclusions & Future Work

This chapter presents the conclusion of the proposed system, summarizing the main findings and overall performance. It also outlines potential directions for future work and possible improvements.

5.1 Conclusions

This thesis addressed the challenge of improving face recognition performance under uncontrolled conditions, where factors such as pose variation, occlusion, and appearance changes can significantly impact accuracy. To tackle this problem, a face recognition system was designed and implemented using a combination of deep learning-based feature extraction and machine learning classification within a Design Science Research framework.

The proposed system integrates multiple components, including data augmentation, embedding generation using FaceNet, classification through a Support Vector Machine, and a hybrid decision strategy that combines classification with

similarity-based verification. This approach was developed to enhance robustness and improve the system’s ability to generalize beyond strictly controlled datasets.

The evaluation results demonstrate that the system achieved high classification performance on the test dataset, with perfect accuracy, precision, recall, and F1-score. These findings indicate that the combination of feature extraction and classification techniques is effective in distinguishing between different identities under the tested conditions. In addition, the system successfully handled both known and unknown users, highlighting its potential applicability in biometric verification scenarios.

Overall, this research contributes a practical and effective approach to face recognition by addressing key challenges associated with real-world variability. The integration of multiple techniques within a unified system demonstrates how combining methods can improve reliability and decision-making in face recognition applications.

5.2 Future Work

While the proposed system demonstrates strong performance, several opportunities exist for further improvement and extension. One important direction for future work is the use of larger and more diverse datasets that include variations in lighting conditions, backgrounds, and image quality. This would enable a more comprehensive evaluation of the system’s robustness and generalization capabilities in real-world environments.

Another area for enhancement involves improving the scalability of the system.

Currently, the need to retrain the model when new users are added can be time-consuming. Future work could explore incremental learning or online learning techniques to allow the system to adapt more efficiently without full retraining. Further improvements could also focus on optimizing the decision-making process, such as dynamically adjusting thresholds based on environmental conditions or confidence levels. This would enhance the system’s adaptability across different use cases.

Finally, deploying the system in real-world environments and conducting user-based evaluations would provide valuable insights into its practical usability, performance, and limitations. Such studies would help bridge the gap between experimental results and real-world applications.

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